

Quantum Oracle Sketching

A method that lets a **quantum computer read classical data in a fundamentally quantum way**. A small QC can **process massive datasets** by examining them one by one.

Usefulness

This eliminates the need to store an entire dataset in memory all at once

Oracle: “black box function”, a hidden mathematical lockbox that EC

--

Streaming Data

Streaming refers to the **continuous**, real-time transmission and **processing of quantum information**.

Useful

enables an ongoing “pipe” of data between standard computers and quantum processors, allowing for continuous computation, feedback, and error correction.

--

Feature Vector

Translates classical data (like a regular list of numbers, images, traditional code bits) **into a quantum state**.

Useful

Allowing for quantum computers to analyze and recognize patterns in information using unique mathematical properties of quantum mechanics

The technique can lead to **quantum feature maps** and **quantum embedding**, which acts like a superpower for machine learning by letting computers process data in massive dimensions

Ex: [25, 50k] This is a feature vector that represents someone's age and yearly income

--

K-mer

A k-mer is a short, **fixed-length sequence** of characters where K = length of the string

Ex: if string is AATTGGCG and k = 3, the possible 3-mers are AAT, ATT, TTG, and so on

--

PCA (principal component Analysis)

A quantum algorithm that **simplifies massive, complex datasets** to find **the most important patterns**

Useful

Allows datasets to shrink by looking at what matters most

- - - - -

Fingerprint

A technique that turns massive amounts of classical data into a much smaller and compressed “signature” encoded on quantum particles.

Allows scientists to compare files using a fraction of the data and energy

Quantum fingerprinting in 3 main steps

Data compression: fingerprints into few qubits

Superposition: because qubits can be in superposition they can represent the entire data set in a compact state

Swap test: test instantly reveals if the fingerprints are an exact match

Usefulness

Enhanced privacy because not a lot of excess data is being shared when fingerprinting is done

- - - - -

Verification

The process of proving that a quantum computer performs a calculation correctly

****Superposition causes this to be hard since it can be in complex probabilistic states**

How to verify

1. Trap/Blind computations give the QC trap question sthat you know the answer to, if the QC answers correctly you know that its other calculations are correct too.
2. Mathematical proof
3. On-chip self-testing